

SANTHOSH REDDY KATASANI VENKATA

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EDUCATION

University of Illinois Chicago, IL, USA

May 2026

Master of Science in Computer Science

Coursework - Neural Networks, Advanced Machine Learning, ML on Graphs, Deep Learning for Computer Vision, Visual Data Science, Natural Language Processing, Pattern Recognition, Big Data Mining, Computer Algorithms

SRM Institute of Science and Technology, Chennai, India

May 2024

Bachelor of Technology in Computer Science

EXPERIENCE

Research Assistant - UI Health, Chicago, USA

June 2025 - Present

- Designed and developed an AI-powered nutrition recommendation platform using Retrieval-Augmented Generation (RAG) and OpenAI LLMs to generate personalized cancer-friendly meal plans based on calories, cuisine, and dietary restrictions.
- Built scalable backend APIs using FastAPI and integrated FAISS vector search to enable fast and relevant retrieval from structured nutrition knowledge sources.
- Developed an interactive React frontend and supported Azure-based deployment for secure, scalable, production-ready access.

Software Development Intern - Soft Logic Systems, Chennai, India

June 2023 - Nov 2023

- Contributed to backend application development using Python and Java for enterprise software solutions, integrating REST APIs and MySQL databases to support business workflows and reliable data exchange.
- Assisted in debugging, feature testing, and Agile sprint delivery, helping improve code quality, system stability, and overall development efficiency.

PROJECTS

Graph Attention Networks for Audio Interference Reduction

April 2026

- Developed three ML pipelines for 3-microphone / 3-source audio interference reduction using raw waveform, spectrogram, and graph-based frequency-domain representations in simulated room acoustics.
- Built Graph Attention Network (GAT) models in PyTorch with CNN encoder-decoder layers, spectral mask prediction, STFT/iSTFT reconstruction, multi-head attention, residual gating, and training optimization for multi-channel source separation.
- Designed a novel frequency-bin graph model treating spectrogram bins as nodes, achieving best performance by capturing inter-frequency dependencies, local spectral structure, and cross-channel correlations.

Technologies: Python, PyTorch, NumPy | *Architecture:* GAT, CNN Encoder-Decoder, Spectrogram Learning, STFT/iSTFT, SI-SDR

Distribution - Aware Visual Prompt Tuning for Lung Cancer Risk Prediction

Dec 2025

- Developed a deep learning pipeline using Vision Transformers (ViT-B/14) to predict 1-year lung cancer risk from CT scans.
- Applied Distribution-Aware Visual Prompt Tuning on frozen pre-trained medical foundation models for efficient fine-tuning, faster training, improved generalization, and reduced computational cost
- Improved AUC-ROC score from **0.65** - **0.77** using positional encoding, attention pooling, and optimized training strategies.

Technologies: Python, PyTorch, HuggingFace, MLflow | *Architecture:* Vision Transformers, Prompt Tuning, Attention-Based Pooling

Cancer Nutrition Analytics Dashboard

Dec 2025

- Engineered data pipelines using NHANES and BRFSS datasets to analyze nutrition, demographic, and health behavior patterns.
- Performed large-scale data cleaning, transformation, and aggregation using Python, Pandas, and SQL for reliable analytics.
- Built dashboards and analytical reports to uncover insights on cancer nutrition trends, regional risks, and population behavior.

Technologies: Python, Pandas, SQL, D3.js | *Architecture:* Data Pipelines, Analytical Modeling

Neural Machine Translation

May 2025

- Built Seq2Seq LSTM and Transformer-based models for French-to-English neural machine translation, implementing preprocessing pipelines with tokenization, padding, vocabulary indexing, and teacher forcing.
- Evaluated translation quality using BLEU score and improved fluency through attention mechanisms and decoding optimization.
- Developed an end-to-end training and inference pipeline for generating accurate and context-aware translated sentences.

Technologies: Python, TensorFlow/Keras, NLTK | *Architecture:* Encoder-Decoder, LSTM, Transformer, Attention

TECHNICAL PROFICIENCIES

Programming Languages:

Python, Java, HTML, SQL

Machine Learning / Deep Learning:

PyTorch, TensorFlow, Keras, Scikit-learn, Transformers, ViTs, GNNs, CNNs, LSTMs

AI / ML Systems:

OpenAI APIs, Llama, RAG, FastAPI, HuggingFace

Data Engineering / Analytics:

Pandas, NumPy, PySpark, SQL, Data Cleaning, Feature Engineering, Dashboards

Deployment / Tools:

Docker, Azure, REST APIs, React, Git, Linux, Jupyter, VS Code, Eclipse